Leonardo Souza

Backend, Web & AI

São Paulo, BR · leonardosouzasilva9@gmail.com · github.com/souzxxx · linkedin.com/in/leonardo-souzx

SUMMARY

Computer Science student at Insper (fifth semester). I work on backend, web and applied AI: an LLM assistant over real user data, vector search in Postgres with pgvector, an outbox queue with exponential retry and idempotency, rate limiting in Redis with a circuit breaker, and a Next.js/TypeScript front-end in a system with real users. I write integration tests against Postgres and Redis, not mocks.

PROJECTS

Transactional e-commerce (under NDA) 2026 · private repository

Java 21 · Spring Boot 4.1 · PostgreSQL · Redis · Next.js · Testcontainers · Flyway

- Own store for a Brazilian brand — name under contract — modular monolith of 12 domains with a Next.js storefront.
- Transactional outbox for email: claim in a short transaction, send outside the lock, and exponential backoff from 30s doubling to 1h, with jitter.
- End-to-end idempotency: payment webhook validated by signature and deduplicated; Idempotency-Key on the email provider.
- Per-key rate limit in Redis via an atomic Lua script that fails open, with a circuit breaker and half-open probe.
- Flyway migrations and 218 test files — 139 on the backend, with Testcontainers bringing up real Postgres and Redis, and 79 on the frontend.

FinanceHub 2026 · financehub-web-ecru.vercel.app

Next.js · TypeScript · Python/FastAPI · Supabase · PostgreSQL · Groq (Llama 3.3)

- Finance platform in production with the Luna assistant: context built from the authenticated user's own transactions, categories and budgets, through a parameterized query filtered by user_id.
- Cost per request kept in check: a 500-token output cap, history truncated to 10 messages, and per-route rate limiting with slowapi (30 req/h on chat, 10/h on insights, 5/h on the report).
- Row Level Security in Postgres, on the digest tables; Turbo monorepo (FastAPI + Next.js) with continuous deploys on Vercel.

Project Wraeclast 2026 · github.com/souzxxx/project-wraeclast

Python · FastAPI · PostgreSQL · pgvector · Next.js · GitHub Actions

- RAG assistant: vector search in Postgres with pgvector, 1024-dimension embeddings and an HNSW index on cosine distance — no fine-tuning, with the LLM answer restricted to the retrieved context.
- Daily data collection in GitHub Actions, kept out of the serverless API by the execution time cap; 350 pytest tests and CI of ruff + pytest on every push.

Sentinel 2026 · github.com/souzxxx/sentinel

Next.js 16 · React Three Fiber · GLSL · Python/FastAPI · WebSocket · Docker

- 3D monitoring dashboard: a FastAPI backend pushes CPU, RAM and disk metrics over WebSocket, with automatic reconnection, and custom GLSL shaders driven by them.

EDUCATION

Insper — B.S. in Computer Science, fifth semester (in progress, 2026).

Current semester: Platforms, Microservices and APIs; AI Startup; Big Data; Algorithm Analysis and Technical Interviews; Games and Interaction.

Earlier: Software Design and Agile Management; Machine Learning; Artificial Intelligence and Robotics.

STACK

Languages TypeScript, Python, Java, JavaScript, C, C# (Unity), Prolog

Backend & web Node.js, Next.js, React, FastAPI, Spring Boot, REST

Data & AI PostgreSQL, pgvector, Redis, Supabase, RAG and embeddings

Infra & quality Docker, Vercel, Railway, GitHub Actions, Testcontainers, Flyway